TCR series

Industrial Robot

TCR10LA/TS3100

INSTRUCTION MANUAL

TRAVERSE AXIS
TRANSPORTATION AND INSTALLATION MANUAL

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

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Shibaura Machine

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CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger or caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

! CAUTION

- Be sure to read through the Safety Manual provided separately before starting actual work to assure safety of the work covering from the robot installation to operation.

[Installation and transportation]

Be sure to observe the following items to use the robot safely.

! DANGER	
 Prohibited	• DO NOT install or operate if any parts are damaged or missing. Otherwise, electric shocks, fires or faults could be caused. • DO NOT install the robot where it may be exposed to splash of liquids such as water. Otherwise, electric shocks, fires or faults could be caused. • DO NOT place the robot near the combustibles. Otherwise, fires could be caused if the material ignites due to a fault, etc.
 Mandatory	• Be sure to secure the robot with the attached clamps before carrying the robot. Otherwise, you could be involved in injuries if the arm moves when the robot is lifted. • Connect the robot only after installation. If wiring is performed before the robot installation, you could be involved in electric shocks or injuries. • Be sure to use the master power voltage and power capacity designated by SHIBAURA MACHINE. Otherwise, the equipment could be damaged or catch fires. • Be sure to use the designated power cable. Otherwise, fires or faults could be caused.
 Always ground	• Completely connect the grounding cable. Otherwise, electric shocks or fires could be caused if a fault or leak current occurs, in addition to malfunction by noise.

! CAUTION	
 Prohibited	• NEVER lift the robot arm during the transport. Otherwise, an excessive force will be exerted on the mechanical part of the robot, resulting in faults. • For the controller, keep an ample space for air vent. Otherwise, the controller will be heated, causing faults.
 Mandatory	• When lifting up the robot, do so carefully. If lifted suddenly, the robot will tilt, which is very dangerous. • When storing the robot, completely clamp the base. If placed directly on the floor, the robot will be unstable and fall over.

Table of Contents

Page

1. Transportation 7
   1.1 Mass and Dimensions 7
   1.2 Packaging Posture of the Robot 7
   1.3 Transporting the Robot 8

2. Installation 10
   2.1 Installing the Robot 10
   2.2 Assembling Hand 11
   2.3 Installing Controller 11

3. Connection 12
   3.1 Cables between Robot and Controller 13
   3.2 External I/O Signal Cable 14
   3.3 Extension I/O Signal (Distribution I/O) Cable 14
   3.4 Power Cable 14
   3.5 Hand Signal Cable 14
   3.6 Hand Piping 14

1. Transportation

1.1 Mass and Dimensions

Operating range of traverse axis	700 mm
Mass of traverse axis	220 kg
Linkage or non-linkage of traverse axis	Non-linkage
Total mass	400 kg

For details on the outer dimensions, see the specifications manual provided separately.

1.2 Packaging Posture of the Robot

The robot is factory-packaged as shown below. To transport the robot, insert the fork of a forklifter into the space for lifter of the wainscot board.

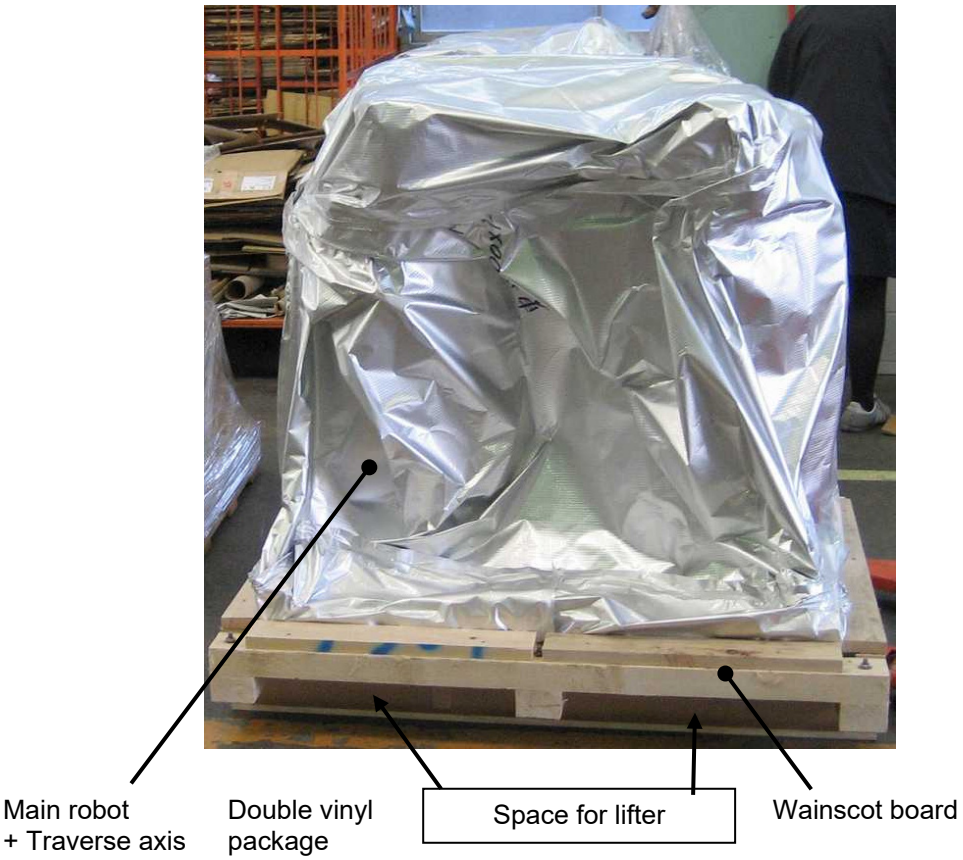


Fig. 1.1 Packaging posture of the robot

1.3 Transporting the Robot

Before transporting the robot, be sure to confirm the following transport procedures and cautions.

! DANGER

- When carrying the robot, be sure to use the attached lifting fixtures.
- Before transporting the robot, be sure to secure the robot with the attached clamps. Failure to do so could cause a hazardous situation as the arm and slider will move when the robot is lifted.

To lift the robot, observe the following procedures.

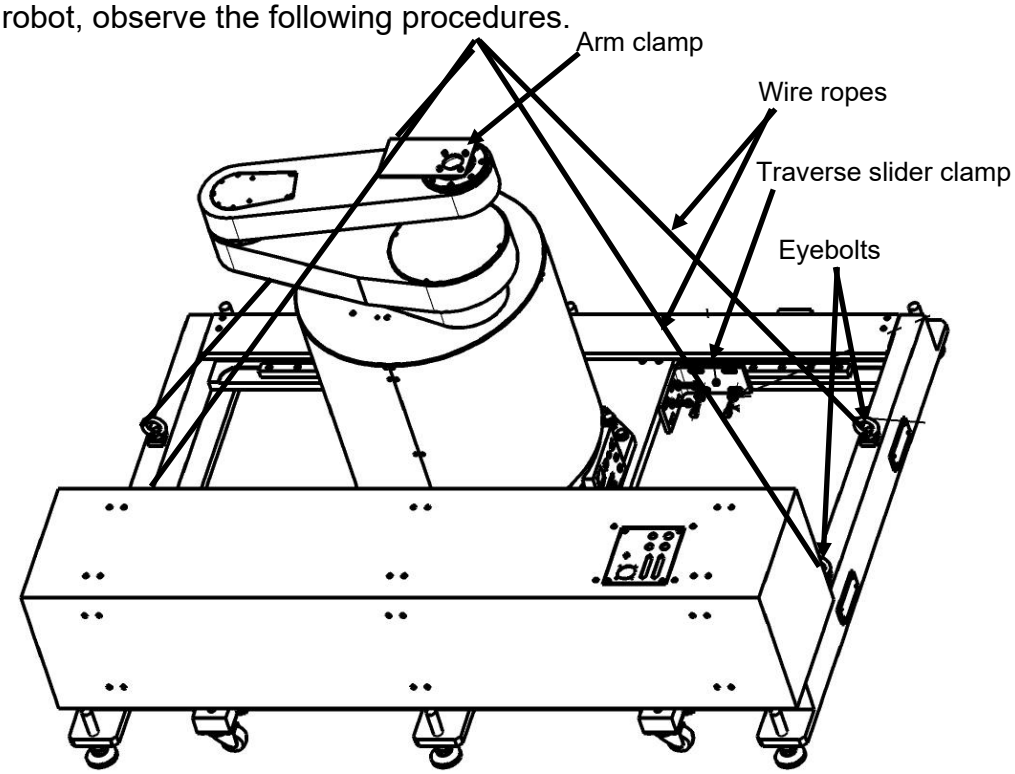


Fig. 1.2 Lifting posture

1) Make sure that the arm is secured with a clamp.
2) Make sure that the traverse slider is secured with a clamp.
3) Attach the M16 eyebolts to the screw holes on the traverse axis frame.

4) Pass the wire ropes to the eyebolts. To assure safety, be sure to use the four (4) wire ropes.
   If the wire ropes may come into contact with the main robot due to the rope length, use cushion material.

5) Lift the robot carefully, using a crane.

! CAUTION

- Use a wire with a limit load that can sufficiently withstand the robot mass.
- The robot will tilt somewhat when lifted. Lift it up carefully and slowly.
- When raising and lowering the robot, DO NOT exert unusually large impact (9.8 m/s^2 or over) on it. Carefully handle the robot.
- When placing the robot on the floor, take careful precautions to prevent your hand or leg from being caught in the robot.

When using the lifter, follow the procedures below.

1) Make sure that the traverse slider is present.

2) Make sure that the traverse slider is secured with the clamp.

3) Jack up the traverse slider, using the M20 taps on the traverse axis frame, until it reaches the height where the fork can be inserted to under the frame.

4) Insert the fork under the traverse axis frame, then carefully lift up the robot by means of the forklifter.

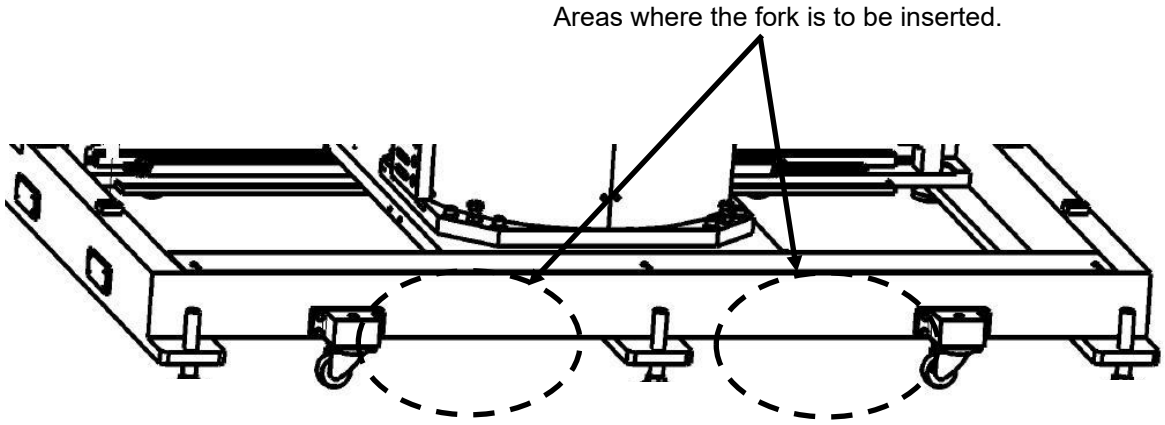


Fig. 1.3 Transporting posture

2. Installation

2.1 Installing the Robot

When installing the robot, strictly observe the following installation procedures and precautions.

1) Temporarily place the robot at a place where it is to be installed.
2) Adjust the jacks with leveling bolts, etc., using the M20 taps on the traverse frame, then perform temporary leveling-off for the robot.
3) Remove the traverse slider clamp. Now, the traverse slider can be moved by hand.
4) Place level gages on the traverse base, then perform leveling off by moving the traverse slider by hand. Use the level gages whose sensitivity is 0.05 mm per 1000 mm.

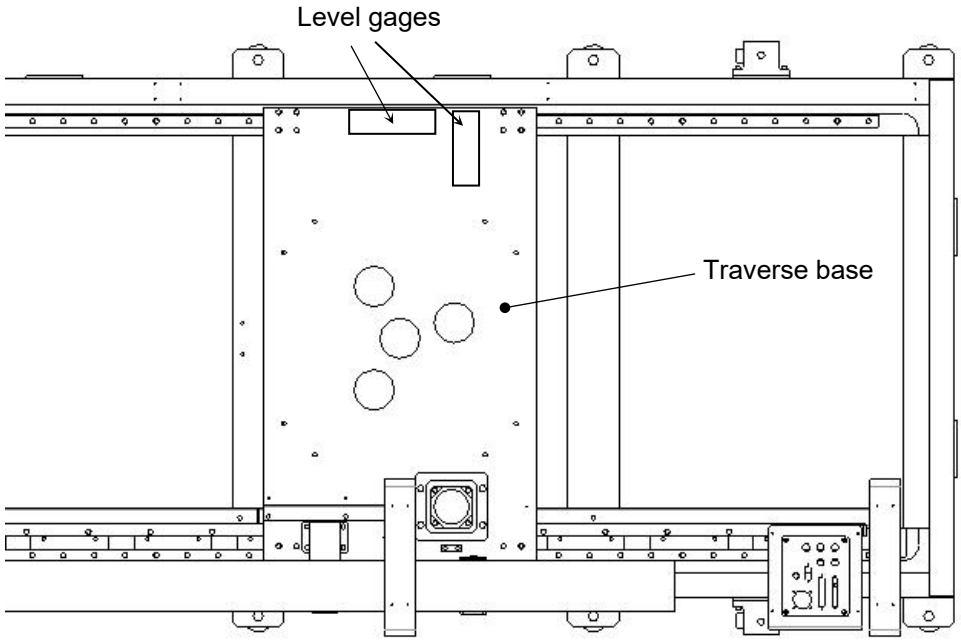


Fig. 2.1 Leveling off the robot

5) When the robot has been leveled off, secure it on the equipment side.
6) Move the robot by hand over the full stroke and make sure that it can be moved smoothly over the full stroke.
7) Remove the arm clamp. Keep the clamp not to miss.

Now, the robot has been installed.

2.2 Assembling Hand

Assemble the robot hand manufactured and provided by the customer. For details, see the specifications manual provided separately.

2.3 Installing Controller

Install the controller. For details, see Section 2 of the Transportation and Installation Manual (STE 80856) provided separately.

3. Connection

Perform wiring and piping for the robot, as shown below.

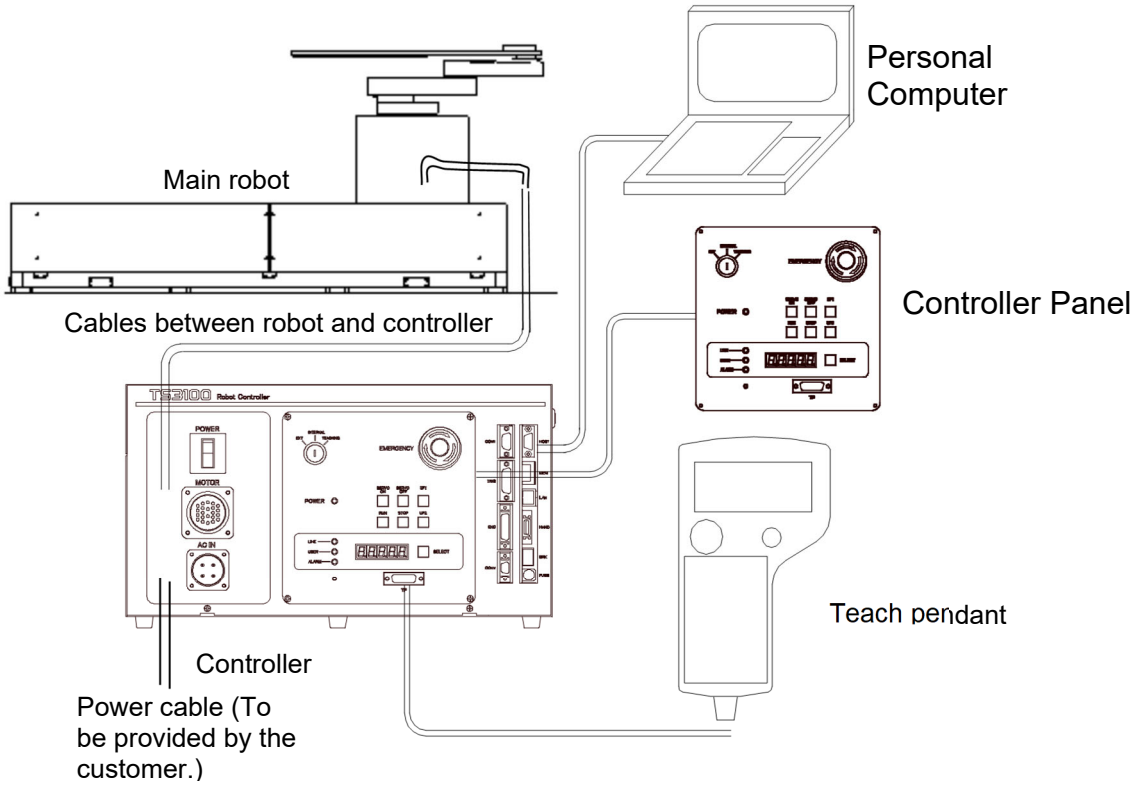


Fig. 3.1 Wiring for the robot

3.1 Cables between Robot and Controller

Connect the cables between the robot and controller. Connect the motor cable, encoder cable and brake cable. For details, see Section 3 of the Transportation and Installation Manual (STE 80856) provided separately.

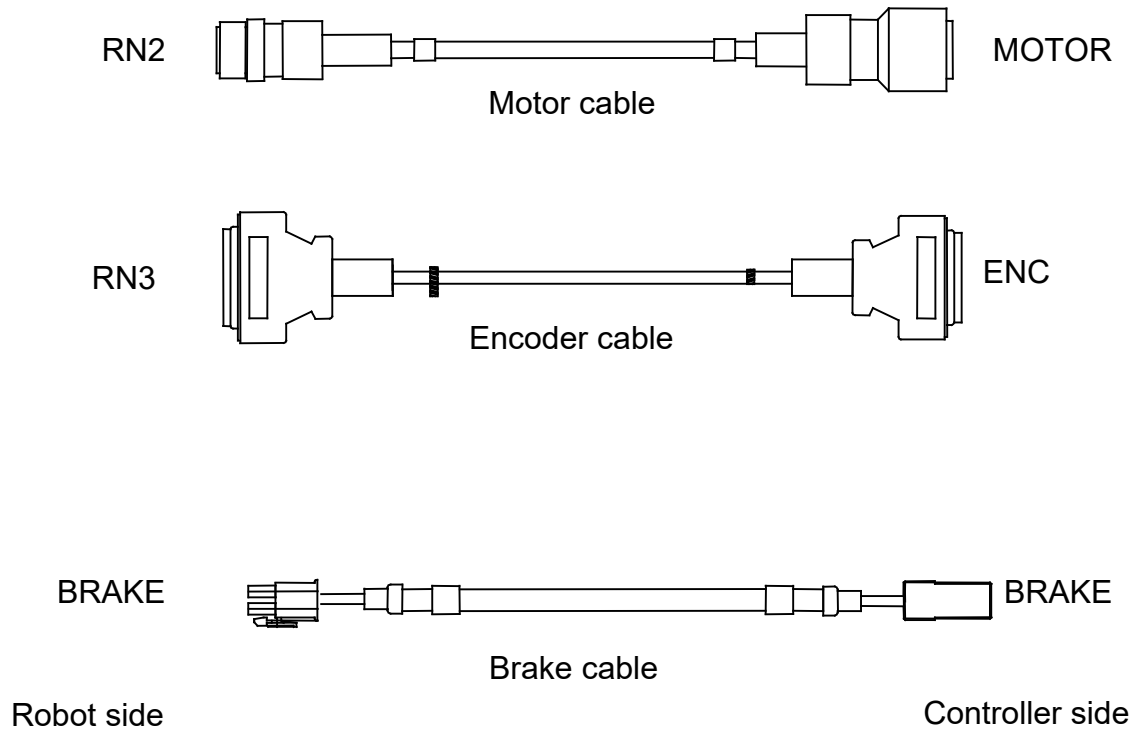


Fig. 3.2 Connection of cables between robot and controller

To control the I/O signals for the hand sensor and solenoid valve from the controller, connect the robot control signal cable. CN11

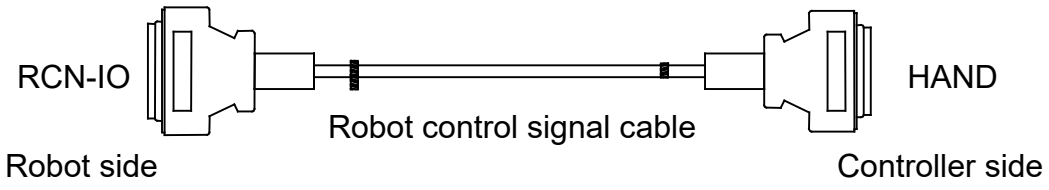


Fig. 3.3 Connection of robot control signal cable between robot and controller

When the I/O signals for the hand sensor and solenoid valve are directly controlled from an external control device, relevant cables between the robot and the same device should be manufactured and provided by the customer. For details, see the specifications manual provided separately.

3.2 External I/O Signal Cable

When using the external I/O signals, connect external I/O signal cables INPUT, OUTPUT. These cables should be manufactured and provided by the customer. For details, see Section 3 of the Transportation and Installation Manual (STE 80856) provided separately.

3.3 Extension I/O Signal (Distribution I/O) Cable

When using the extension I/O signals (distribution I/Os), connect extension I/O signal (I/O distribution) cables. These cables should be manufactured and provided by the customer. For details, see Section 3 of the Transportation and Installation Manual (STE 80856) provided separately.

3.4 Power Cable

Connect power cable CN1. This cable should be manufactured and provided by the customer. For details, see Section 3 of the Transportation and Installation Manual (STE 80856) provided separately.

3.5 Hand Signal Cable

Connect the hand signal cables with the hand sensor. Terminal treatment for the sensor should be done by the customer. For details, see the specifications manual provided separately.

3.6 Hand Piping

Perform primary piping from the traverse axis connector panel and piping on the hand side from the end effector. For details, see the specifications manual provided separately.

Now, the connection completes.
When confirming the robot motions finally, be sure to move the robot at a low speed and by jog operation.